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A 2D Peridynamic Model for Failure Analysis of Orthotropic Thin Plates Due to Bending

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Abstract

The present paper deals with the problem of cracks propagating in thin orthotropic flat plates under bending loads. We define the mechanical behavior of an orthotropic lamina in which shear deformation can be neglected (Kirchhoff plate). The formulation follows the main ideas of bond based Peridynamics. Failure criteria limit the maximum curvatures and provide failure values which may be different in different directions. Several numerical examples show that the results obtained with the new approach are in good agreement with those obtained with more classical computational methods. Moreover the numerically computed crack patterns seem to follow in a reasonable way the orthotropic properties of the models.

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Keywords: Peridynamic Theory; Bending Failure

Nomenclature

PD	Peridynamics
CCM	Classical Continuum Mechanics
CLT	Classical Laminate Theory
SED	Strain Energy Density
FEA	Finite Element Analysis

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1. Introduction

Many aerospace structures use composite materials with orthotropic behavior to take advantage of their specific stiffness and strength. However the failure prediction methods for composite materials are not well established as those for metallic materials in classical continuum mechanics. Thus structural justification of composite structures is mostly based on experimental data. The failure behavior of composite materials can be extremely articulated: the dynamic loading may give rise to fracture behavior which is different from that of quasi-static conditions (Kazemahvazi et al., 2009). The splitting mode (debonding between matrix and fibers) and matrix cracking are the most common intralaminar fracture mode in unidirectional fiber reinforced materials under quasi static load conditions. Extensive fracture and damage produced by interconnected splitting, matrix cracking, delamination and fiber breakage, are observed under dynamic loading conditions (Haque, 2005).

In classical continuum mechanics (CCM) modeling of crack initiation and propagation is not possible without modifications to its original formulation. The difficulty of crack modeling in CCM comes from its governing equations which require spatial derivatives. Spatial derivatives are not defined at crack tips by definition. As a remedy, Silling (2000) proposed a nonlocal theory of continuum mechanics called Peridynamics (PD). PD employs integro-differential equations which remains valid at discontinuities. Thus, a special treatment at the crack tip is not necessary and crack initiation and crack growth can be analyzed without requiring an external criteria (Oterkus and Madenci, 2012).

Two main approaches have been identified to apply the PD theory to the study of composite materials: one based on homogenized models and one based on the explicit model of fiber and matrix materials. A first example of peridynamic model used to study fracture and damage of composite materials is presented in Hu et al. (2011) and Hu et al. (2012) to analyze the damage patterns in laminated composites subjected to bi-axial loading. Ghajari et al. (2014) derived the PD material bond parameters as continuous functions of bond orientation and they validated their model results for uniaxial tension of composite plate and Cortical bone Compact Tension specimen. The homogenized PD model has been also extended to take into account the transverse elastic module of the lamina, using a 3D approach (Hu et al., 2014).

The explicit model of fibers (or fibers group) and matrix, able to capture the splitting fracture mode, is presented in Kilic et al. (2009). Explicit modeling has the advantage of obtaining the most detailed possible solution, but demands higher computational cost. However this last issue could be solved using an adaptive meshing approach (Dipasquale and Zaccariotto, 2014). Current peridynamic models capture tension and compression in 2D membranes, but these do not resist to transverse loads, in this case a complete 3-dimensional solid model should be used.

A recent paper reduces a bond based 3D plate to two dimensions with an integral through the plate's thickness (Taylor and Steigmann, 2013). This creates a model that can represent thin structures and includes a bending term, but additional equations are needed. O'Grady and Foster (2014) use the state based PD (that requires more computational effort with respect to the bond based version) and instead of build up new equations of motion they develop a new material constitutive model (non-ordinary bond-pair model). Finally Diyaroglu et al. (2015) proposed a Mindlin Plate PD formulation which also considers transverse shear deformations.

In this paper a peridynamic model for orthotropic plates, using the homogenized approach, will be proposed and applied to static and dynamic analyses of progressive damage. The new model will take into account the bending stiffness in the 2D formulation extending the approach described in Diyaroglu et al. (2015) to orthotropic materials.

2. Theory

2.1. Peridynamic Theory

In (CCM), deformation of a body subjected to external loads can be calculated by treating the body as a continuum. In CCM, it is assumed that a continuous medium has infinite number of infinitesimal volumes which interact only with their immediate neighbors. Equation of motion in CCM can be written as:

$$-\rho(\mathbf{x})\ddot{\mathbf{u}} + \nabla \cdot \boldsymbol{\sigma} + \mathbf{b}(\mathbf{x}, t) = 0. \quad (1)$$

where $\rho(\mathbf{x})$, $\ddot{\mathbf{u}}$ and $\mathbf{b}(\mathbf{x}, t)$ denote mass density, spatial acceleration and body force density vector of the material point $\mathbf{x}$, respectively. The spatial derivatives of stress tensor $\boldsymbol{\sigma}$ required in Eq. (1) are not valid at discontinuities. In PD, equation of motion does not require spatial derivatives and can be written as:

$$\rho(\mathbf{x})\ddot{\mathbf{u}}(\mathbf{x}, t) = \int_H \mathbf{f}(\mathbf{u}' - \mathbf{u}, \mathbf{x}' - \mathbf{x}) dH + \mathbf{b}(\mathbf{x}, t). \quad (2)$$

In Eq. (2), H is the horizon of material point $\mathbf{x}$ with a radius of δ , $\mathbf{x}'$ is a family material point, ρ is the mass density, $\mathbf{u}$ and $\mathbf{b}$ denotes the displacement and body force vector fields, respectively. Note that $\mathbf{f}(\mathbf{u}(\mathbf{x}') - \mathbf{u}(\mathbf{x}), \mathbf{x}' - \mathbf{x})$ is the pairwise force vector that the material point $\mathbf{x}$ exerts on the material point $\mathbf{x}'$. It has been shown that, PD converges to classical theory of elasticity, when the length of the horizon goes to zero (Silling and Lehoucq, 2008).

2.2. Mindlin Plate Kinematics (Diyaroglu et al., 2015)

As shown in Fig.1, $\phi_{(j)}$ and $\phi_{(k)}$ represent the rotations of material points j and k with respect to line of action between of material points j and k . The curvature between material points k and j $\kappa_{(k)(j)}$ can be written as (Diyaroglu et al., 2015):

$$\kappa_{(k)(j)} = \frac{\phi_{(j)} - \phi_{(k)}}{\xi_{(j)(k)}}. \quad (3)$$

By applying coordinate transformation, the rotations of a two dimensional plate can be written as (Diyaroglu et al., 2015):

$$\phi_{(j)} = \phi_{x(j)} \cos \theta + \phi_{y(j)} \sin \theta, \quad \phi_{(k)} = \phi_{x(k)} \cos \theta + \phi_{y(k)} \sin \theta. \quad (4)$$

where θ is the bond angle with respect to x axis direction (see Fig.1a). The plate curvature $\kappa_{(k)(j)}$ can be rewritten using Eq. (4) as follows:

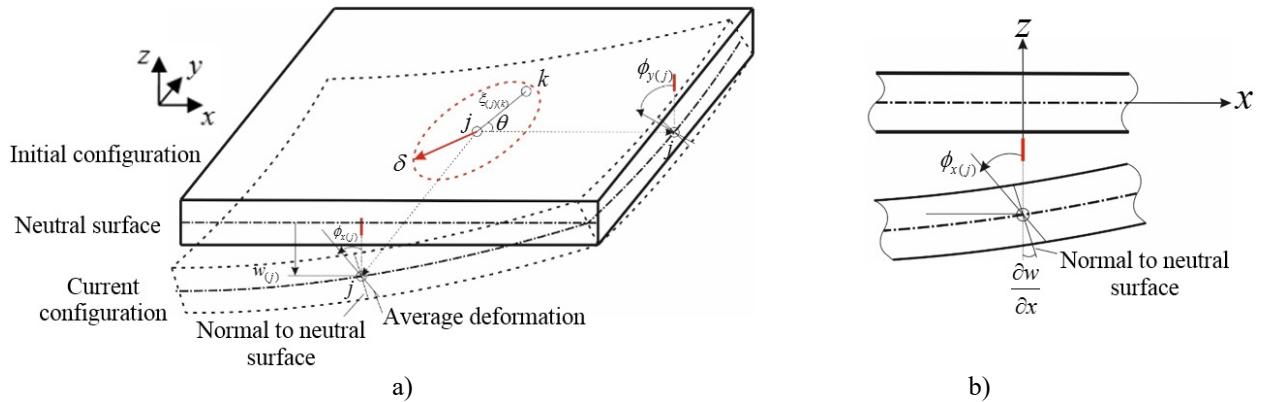


Fig. 1: a) Initial and current configurations of a Mindlin plate b) representation of rotation at material point j (Diyaroglu et al., 2015)

$$\kappa_{(k)(j)} = \left(\frac{\phi_{x(j)} - \phi_{x(k)}}{x_{(j)} - x_{(k)}} \right) \cos^2 \theta + \left(\frac{\phi_{y(j)} - \phi_{y(k)}}{y_{(j)} - y_{(k)}} \right) \sin^2 \theta, \quad (5)$$

where

$$x_{(j)} - x_{(k)} = \xi_{(j)(k)} \cos \theta, \quad y_{(j)} - y_{(k)} = \xi_{(j)(k)} \sin \theta. \quad (6)$$

As illustrated in Fig.1, transverse shear angles at material points, $\varphi_{(j)}$, $\varphi_{(k)}$ and the slope $\vartheta_{(k)(j)}$ can be written as:

$$\varphi_{(j)} = \vartheta_{(k)(j)} - \phi_{(j)}, \quad \varphi_{(k)} = \vartheta_{(k)(j)} - \phi_{(k)}, \quad \vartheta_{(k)(j)} = \frac{w_{(j)} - w_{(k)}}{\xi_{(j)(k)}}. \quad (7)$$

By using Eq. (7), the transverse shear angle between material points j and k can be defined as the average of the transverse shear angles at these material points and can be written as

$$\varphi_{(k)(j)} = \frac{w_{(j)} - w_{(k)}}{\xi_{(j)(k)}} - \frac{\phi_{x(j)} + \phi_{x(k)}}{2} \cos \theta - \frac{\phi_{y(j)} + \phi_{y(k)}}{2} \sin \theta. \quad (8)$$

The equations of motion have been derived in Diyaroglu et al. (2015) using the principle of virtual work and solving the Lagrange equations. The obtained formulas are:

$$\rho h \ddot{w}_{(k)} = c_s \sum_{j=1}^{\infty} \varphi_{(k)(j)} V_{(j)} + \hat{b}_{(k)}, \quad (9a)$$

$$\rho \frac{h^3}{12} \ddot{\phi}_{x(k)} = c_b \sum_{j=1}^{\infty} \kappa_{(k)(j)} \cos \theta V_{(j)} + \frac{1}{2} c_s \sum_{j=1}^{\infty} \xi_{jk} \varphi_{(k)(j)} \cos \theta V_{(j)} + \tilde{b}_{x(k)}, \quad (9b)$$

$$\rho \frac{h^3}{12} \ddot{\phi}_{y(k)} = c_b \sum_{j=1}^{\infty} \kappa_{(k)(j)} \sin \theta V_{(j)} + \frac{1}{2} c_s \sum_{j=1}^{\infty} \xi_{jk} \varphi_{(k)(j)} \sin \theta V_{(j)} + \tilde{b}_{y(k)}. \quad (9c)$$

In Eq. (9), c_s and c_b are bond parameters related to transverse shear and bending deformations respectively. Those parameters are given by Diyaroglu et al. (2015):

$$c_s = \frac{9E}{4\pi\delta^3} k^2, \quad c_b = \frac{E}{\pi\delta} \left(\frac{3h^2}{4\delta^2} + \frac{27}{80} k^2 \right). \quad (10)$$

2.3. Orthotropic Plate

For orthotropic plates, Classical Laminate Theory (CLT) is commonly used in practical applications. CLT is an extension of Kirchhoff's Plate Theory. In CLT it is assumed that sections always stay perpendicular to the bending plane. The Mindlin Plate formulation in Diyaroglu et al. (2015) is modified with the assumption no shear deformation. This means that the shear angles $\varphi_{(j)}$ and $\varphi_{(k)}$ defined in Eq. (7) are equal to zero and then also the transverse shear angle between material point k and j $\varphi_{(k)(j)}$ is equal to zero. So the motion equations Eq. (9b) and Eq. (9c) can be re-written introducing also the angle dependence of micromodulus c in the following terms:

$$\rho \frac{h^3}{12} \ddot{\phi}_{x(k)} = c(\theta) \sum_{j=1}^{\infty} \left[\left(\frac{\phi_{x(j)} - \phi_{x(k)}}{x_{(j)} - x_{(k)}} \right) \cos^2 \theta + \left(\frac{\phi_{y(j)} - \phi_{y(k)}}{y_{(j)} - y_{(k)}} \right) \sin^2 \theta \right] \cos \theta V_{(j)} + \tilde{b}_{x(k)}, \quad (11a)$$

$$\rho \frac{h^3}{12} \ddot{\phi}_{y(k)} = c(\theta) \sum_{j=1}^{\infty} \left[\left(\frac{\phi_{x(j)} - \phi_{x(k)}}{x_{(j)} - x_{(k)}} \right) \cos^2 \theta + \left(\frac{\phi_{y(j)} - \phi_{y(k)}}{y_{(j)} - y_{(k)}} \right) \sin^2 \theta \right] \sin \theta V_{(j)} + \tilde{b}_{y(k)}. \quad (11b)$$

As stated in Eq. (11), the bond constant c depends on the orientation of the bond in for orthotropic plates. The dependence of the bond constant on the fiber orientation can be stated as (Oterkus and Madenci, 2012b):

$$c(\theta) = \begin{cases} (c_f + c_m) & \text{if } \theta = \theta_f, \\ c_m & \text{if } \theta \neq \theta_f. \end{cases} \quad (12)$$

Peridynamic moment density $\mathfrak{M}$ and the associated bond strain energy density, $U_{(k)(j)}$ can be defined as:

$$\mathfrak{M}_{(k)(j)}(\kappa_{(k)(j)}) = c(\theta) \kappa_{(k)(j)}(\theta), \quad U_{(k)(j)} = \frac{1}{2} (\mathfrak{M}_{(k)(j)}) \kappa_{(k)(j)}(\theta) \xi = \frac{1}{2} c(\theta) \kappa_{(k)(j)}^2 \xi. \quad (13)$$

where ξ is the bond length. It must be observed that the material point energy density is half of the bond energy density (Hu et al., 2011). The energy density due to bending evaluated at the material point k is

$$U_{b(k)} = \frac{1}{2} \int_H \frac{1}{2} c(\theta) \kappa_{(k)(j)}^2 \xi(\theta) dH, \quad (14)$$

in which $\kappa_{(k)(j)}$ is the curvature along the bond direction.

2.3.1 Orthotropic Plate, strain density energy

The strain energy density due to bending for an orthotropic plate (see Mansfield, 1989 for more details) is reported in the following formula:

$$U_b = \frac{1}{2} \left(D_{11} \kappa_{xx}^2 + 2D_{12} \kappa_{xx} \kappa_{yy} + D_{22} \kappa_{yy}^2 + 2\kappa_{xy} \left(D_{16} \kappa_{xx} + D_{26} \kappa_{yy} + D_{66} \frac{\kappa_{xy}}{2} \right) \right). \quad (15)$$

where the curvatures are defined as

$$\kappa_{xx} = \frac{\partial^2 w}{\partial x^2}, \quad \kappa_{yy} = \frac{\partial^2 w}{\partial y^2}, \quad \kappa_{xy} = 2 \frac{\partial^2 w}{\partial x \partial y}, \quad (\text{Reissner notation for twisting}) \quad (16)$$

In order to find the bond constants, the Strain Energy Density (SED) under the following curvature field will be considered

$$\kappa_{xx} = \zeta, \quad \kappa_{yy} = -\nu_{xy} \zeta, \quad \kappa_{xy} = -\mu_{xy} \zeta. \quad (17)$$

The proposed curvature field represents the one obtained applying a uniform external moment on a plate, M_x . Adopting the discretized form of Eq. (14), the SED in a Peridynamic model can be written as follows:

$$U_{b(k)} = \frac{1}{2} \sum_{q=1}^Q \frac{1}{2} c_f \kappa_{(k)(q)}^2 \xi_{(k)(q)} V_{(q)} + \frac{1}{2} \sum_{j=1}^J \frac{1}{2} c_m \kappa_{(k)(j)}^2 \xi_{(k)(j)} V_{(j)}. \quad (18)$$

In the previous formula Q is the number of fiber bonds while J is the number of matrix bonds within the horizon.

The SED using the CLT can be written as:

$$W_{(k)}^{CLPT} = \frac{1}{2} (D_{11} - 2\nu_{xy} D_{12} - 2\mu_{xy} D_{16} + 2\mu_{xy} \nu_{xy} D_{26} + \nu_{xy}^2 D_{22} + \mu_{xy}^2 D_{66}) \zeta^2. \quad (19)$$

Using Eq. (18) and Eq. (19), bond constants c_f and c_m can be obtained as

$$c_f = \frac{(D_{11} - D_{22})}{\beta}, \quad c_m = \frac{24D_{12}}{\pi h \delta^3}, \quad \beta = \frac{1}{2} \sum_{q=1}^Q \xi_{(q)(k)} V_q. \quad (20)$$

In Eq. (19) and Eq. (20) D_{ij} are the coefficients of the bending stiffness matrix in CLT.

2.4. Failure criteria

Using the proposed Kirchhoff plate bending theory for orthotropic materials and adopting a PD approach it is possible to evaluate the curvature of the plate at each material point. The adopted failure criteria is based on a limit curvature. Following the roadmaps of Hu et al. (2011), the limit curvature could also be direction dependent in the way described in Fig. 2

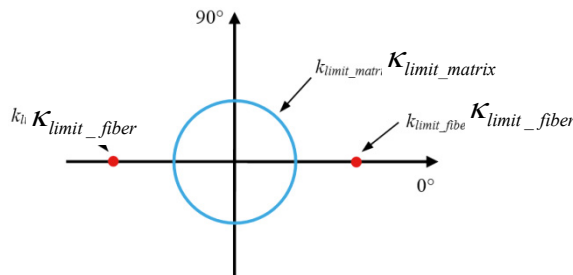


Fig. 2 Variation of limit curvature with respect to the angle between the bond and the fiber direction.

The limit curvature for an isotropic plate is proposed in Diyaroglu et al. (2015) as:

$$\kappa_c = \sqrt{\frac{4G_{IC}}{c_f h \delta^4}}. \quad (21)$$

The extension of the previous formula, for an orthotropic plate is

$$\kappa_{limit_fiber} = \sqrt{\frac{4G_0^{11}}{c_f h \delta^4}}, \quad \kappa_{limit_matrix} = \sqrt{\frac{4G_0^{22}}{c_m h \delta^4}}. \quad (22)$$

κ_{limit_fiber} is the limit curvature for the bond aligned in the “fiber direction” while for the other bonds the limit curvature is κ_{limit_matrix} . The bond constants c_f and c_m appearing in Eq. (22) are defined in Eq. (20). G_0^{11} and G_0^{22} are the fracture energies for a UD lamina in which the axis of the applied moment is parallel and perpendicular to the fiber direction, respectively.

3. Numerical examples

Orthotropic laminas that are considered in this study are square with the side of 1 m ($L = H = 1$ m) and with a thickness of $t = 0.05$ m. The elastic moduli in the fiber and transverse directions of the orthotropic plate are $E_1 = 159.96$ GPa and $E_2 = 8.96$ GPa, respectively, with the usual constraints on the value of the Poisson ratio. The distance between material points is $\Delta x = 0.01$ m in both directions. In order to obtain steady-state solutions, the adaptive dynamic relaxation technique given by Kilic and Madenci (2010) is used.

3.1. Static analyses: linear case

In the first examples, the lamina is clamped at the left edge. This is the reason why the left edge of the lamina is constrained by introducing a fictitious region with a size of $3\Delta x$. All the material points in the fictitious region are fixed. The loading is applied at the right edge as a resultant bending moment of $\tilde{b}_y = 3.33 \times 10^7$ N/m for 0° fiber orientation angle and $\tilde{b}_y = 3.33 \times 10^6$ N/m for 90° fiber orientation angle. Firstly the elastic solutions of a lamina with 0° and 90° fiber orientation as illustrated in Fig. 3 are calculated with the proposed formulation. The PD results are compared with classical FE results obtained using ABAQUS software.

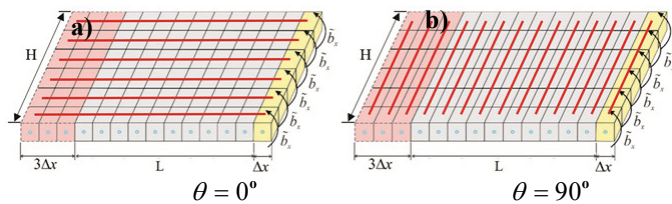


Fig. 3 Orthotropic plates with a) 0° and b) 90° fiber orientations under pure bending moment.

The rotations and deflections with reference to plate length, for the 0° lamina, obtained by PD and FEA solutions, are shown in Fig. 4a and Fig. 4b. The rotations and deflections with respect to plate length, for the 90° lamina, obtained by PD and FEA solutions, are shown in Fig. 4c and Fig. 4d.

We consider next an orthotropic lamina with two different fiber orientations under a pure bending moment load, as shown in Fig. 5. All static examples exhibit a satisfying agreement of the PD solution with the reference solution.

3.2. Dynamics analyses: non-linear case

In the present section we observe as an initial crack in the plate propagates when a bending load is applied. Fig. 7 shows the three different cases under investigation. Then Fig. 8a, Fig. 8b and Fig. 8c present the results about the cases in Fig. 7.a, Fig. 7.b and 7.c respectively. In all cases the crack propagation patterns appear to be reasonable.

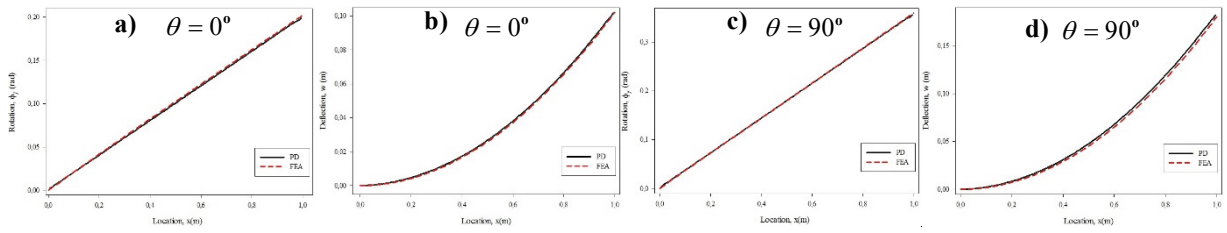


Fig. 4 FEA and PD results comparison for rotations a) and c) and out of plane displacements b) and d) for the orthotropic plates under bending load

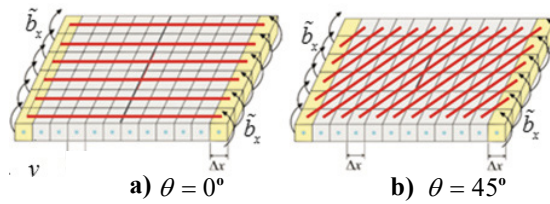


Fig. 5 Orthotropic laminas with a) 0° and b) 45° fiber orientations under pure bending moment conditions.

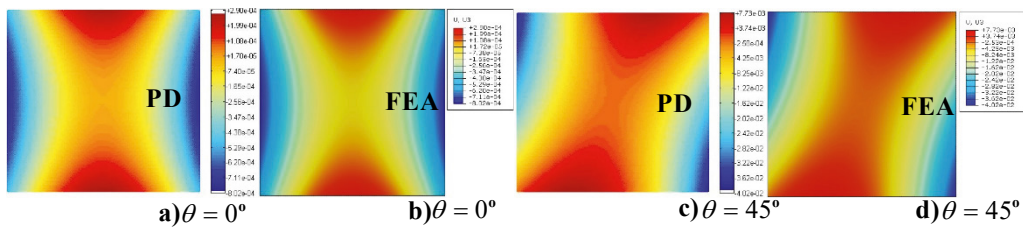


Fig. 6 PD and FEA results comparison for out of plane displacement contours for the orthotropic plates under bending load

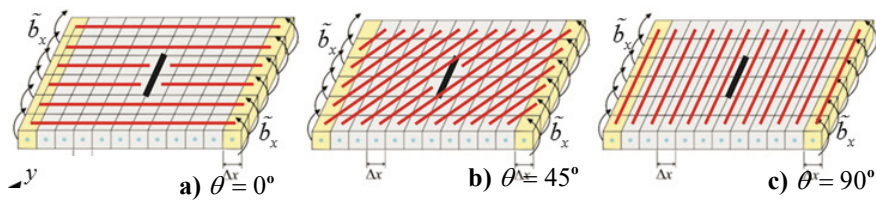


Fig. 7 Orthotropic plates with central crack under pure bending moment

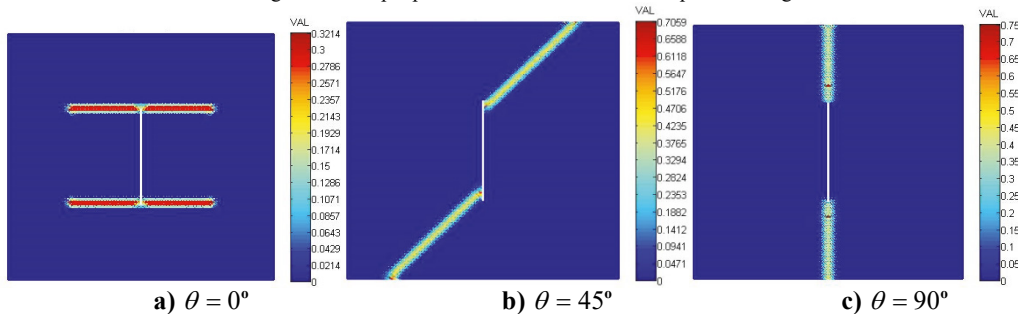


Fig. 8 Damage contours for the orthotropic plates with central crack under bending load.

4. Conclusions

The paper describes a structural model of an anisotropic Kirchhoff plate in the framework of the peridynamic theory. A failure criterion is defined to allow crack propagation. Several examples show the potentialities of the model to reproduce linear elastic results and to simulate dynamic crack propagation in an anisotropic medium. The elastic solutions of lamina under pure bending are in close agreement with FE results. In all dynamic cases the crack propagation patterns appear to be reasonable.

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